# Optimization of the Dominican Republic Energy Matrix: Pathways to Decarbonization (2024-2050)

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**Abstract**

This project models the energy system of the Dominican Republic (DR) to identify optimal pathways for expanding Renewable Energy Sources (RES) and decarbonizing the economy. Utilizing the Open Source Energy Modelling System (OSeMOSYS), the study evaluates the techno-economic feasibility of meeting national targets (Law 57-07) and international commitments (Paris Agreement). This report details the methodology, simulation setup, and a critical analysis of how upgrading input datasets—specifically integrating the transport sector and increasing temporal resolution—impacts long-term energy planning results.

# Contents

# 1 Introduction

## 1.1 Context: Small Island Developing States (SIDS)

The Dominican Republic (DR) is a Small Island Developing State (SIDS) characterized by high economic and geographical vulnerability. Like many SIDS, the DR faces a dual challenge: a high dependency on imported fossil fuels and extreme susceptibility to climate change impacts.

## 1.2 Problem Statement

The primary problem is the need to decouple the DR's economic growth from fossil fuel consumption. The challenge lies in identifying an optimal mix of technologies—Solar, Wind, Biomass, and Hydro—that can meet surging demand without compromising grid reliability or economic viability.

## 1.3 Project Objectives

The objective of this study is to model the DR energy system to analyze cost-optimal pathways for transition. Specifically:

- Recreate the Reference Energy System (RES) of the Dominican Republic using OSeMOSYS.
- Assess the feasibility of meeting Law 57-07 (25% renewable energy contribution).
- Evaluate investment requirements for the Paris Agreement (25% emission reduction by 2030).

# 2 Methodology

## 2.1 The OSeMOSYS Framework

The study utilizes OSeMOSYS, a deterministic linear optimization tool. It minimizes the Total Net Present Cost (NPC) of the system over the study period (2024-2050).

## 2.2 Data Sources and Model Evolution

This project builds upon foundational work (Quevedo et al., 2022) but integrates an updated 2024 dataset. Key enhancements include:

- **Temporal Resolution:** Shift from 12 to 96 time slices (4 seasons $\times$ 24 hours) to capture solar intermittency.
- **Sectoral Coupling:** Integration of the Land Transport sector (EVs, Buses).
- **Updated Emissions:** Use of current IPCC factors.

# 3 Simulation Results & Analysis

## 3.1 Scenario Comparison Overview

The simulation results highlight a distinct divergence between the "Business As Usual" (BAU) path and the transition scenarios (Scenario 1 & 2). Figure 1 provides a comprehensive overview of this divergence across four key metrics.

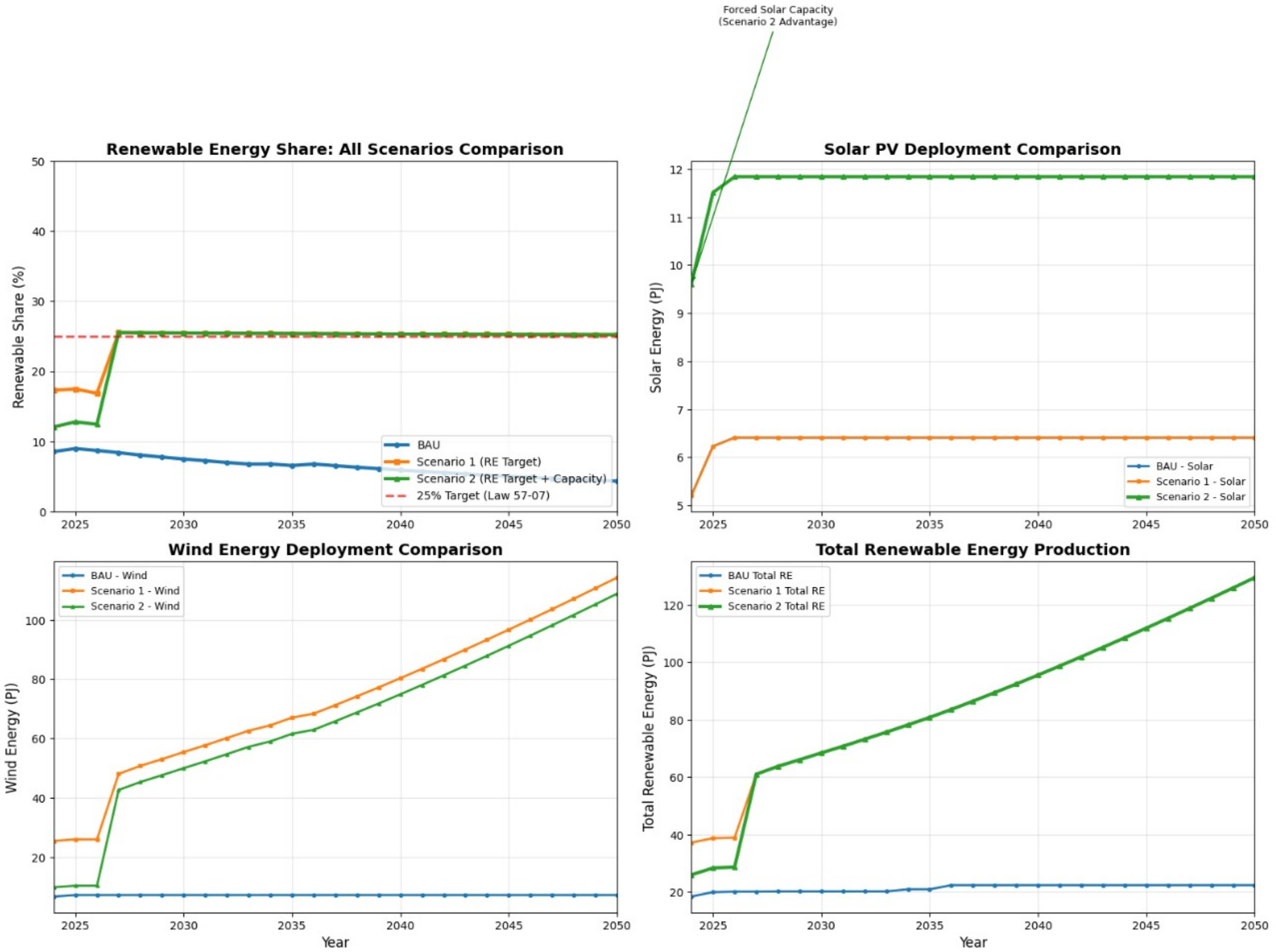


Figure 1: Multi-metric comparison of scenarios. Top-Left: Renewable Energy Share (green line meets target). Top-Right: Solar PV Deployment. Bottom-Left: Wind Energy Deployment. Bottom-Right: Total Renewable Energy Production. Note the sharp increase in Solar and Wind capacity in Scenario 2 compared to BAU.

As seen in the Top-Left panel of Figure 1, the BAU scenario (blue line) stagnates well below the 10% renewable mark. In contrast, Scenario 1 and Scenario 2 (orange and green lines) successfully breach the 25% threshold by 2027, driven by policy constraints.

## 3.2 The Failure of Business As Usual (BAU)

Without policy intervention, the market-optimal solution favors fossil fuels. Figure 2 details the renewable energy profile for the BAU scenario.

The bottom panel of Figure 2 clearly illustrates the "Policy Gap"—the distance between the green line (actual performance) and the red dashed line (legal requirement).

Under the Business As Usual (BAU) scenario, the Dominican Republic's energy system remains largely dependent on fossil fuels throughout the entire planning horizon (2024–2050). Because no renewable-energy mandates or emission-control constraints are imposed, the OSeMOSYS optimization selects the least-cost technologies available — which, given current fuel prices and capital costs, are primarily oil-, coal-, and gas-based generation options. As a result, renewable energy penetration increases only marginally, and the system progresses along a high-emission, fossil-intensive pathway.

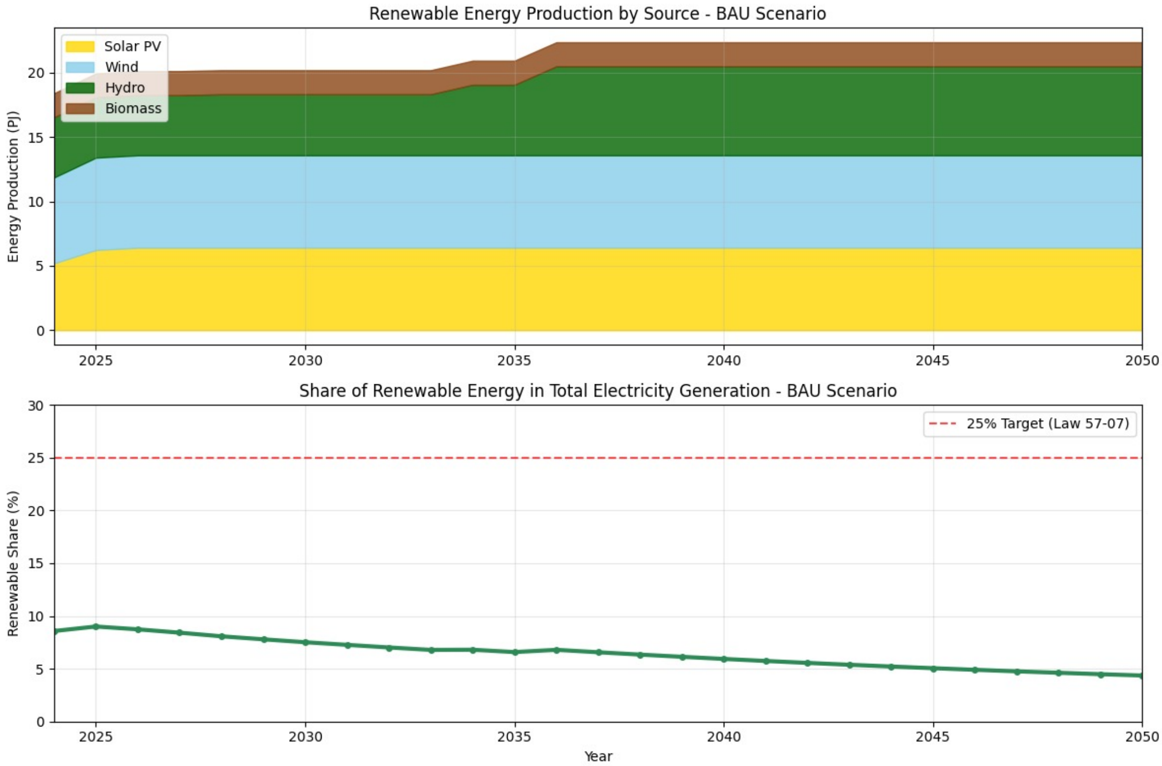


Figure 2: Business As Usual (BAU) Scenario Analysis. Top: Renewable production is dominated by existing Hydro and minor Solar. Bottom: The renewable share (green line) flatlines at ∼9%, failing to meet the 25% Law 57-07 target (red dashed line).

### BAU Energy Production Trend

Figure 3 shows the annual electricity production by technology under BAU. Fossil technologies (Diesel, HFO, LFO, Natural Gas, Coal) dominate the generation mix, occupying the majority of the stacked area. Although some expansion of Solar and Wind occurs, their growth is slow and insufficient to shift the overall composition of the system. Hydro remains constant, limited by resource availability. As electricity demand rises, the additional production is met primarily through increased dispatch of gas and oil technologies rather than renewables.

### BAU Emissions Trajectory

Figure 4 displays the annual emissions by technology for the BAU scenario. Emissions rise steadily over time, largely driven by the increased use of oil- and gas-based technologies to meet growing electricity demand. Wind and Solar provide small emission-free contributions, but their impact is dwarfed by the rapidly increasing fossil-based output. As a result, total emissions surpass 40 Mton $CO_2$-eq by mid-century, placing the country far off track from its Paris Agreement commitments.

The stacked structure of the emissions graph highlights the dominance of high-carbon technologies (HFO, LFO, Diesel, and Coal). Gas contributes a significant but somewhat lower portion of emissions, acting as the system's marginal expansion fuel. The lack of structural policy requirements means that the optimization has no incentive to substitute fossil generation with renewables, and emissions therefore climb monotonically through 2050.

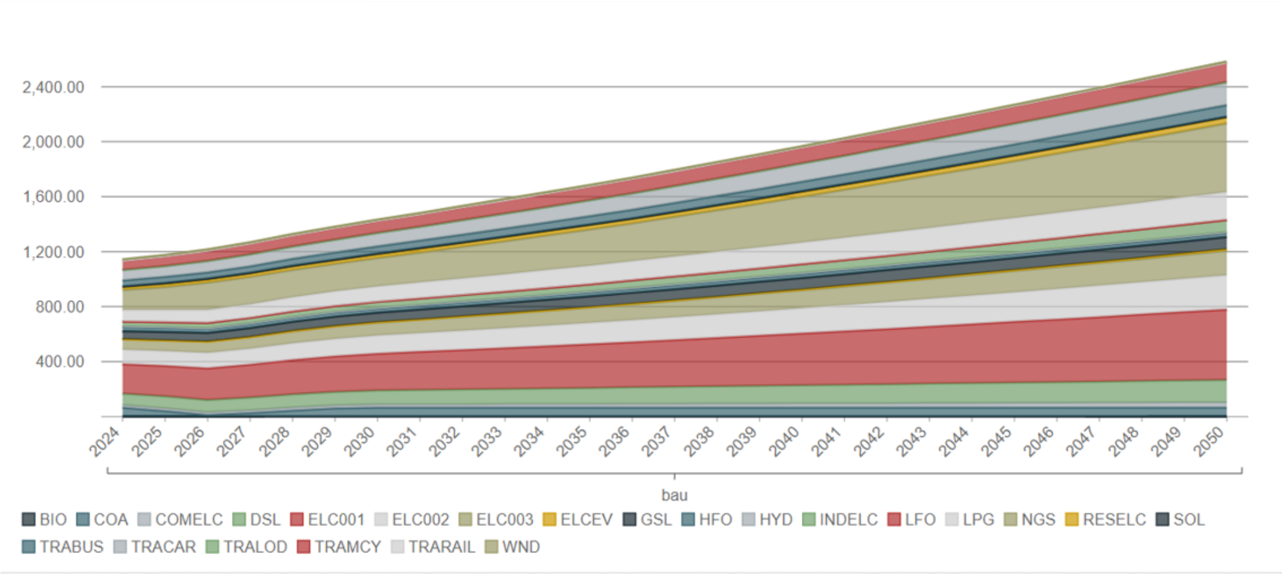


Figure 3: Annual energy production by technology for the BAU scenario (2024–2050). Fossil-fuel technologies dominate the generation profile, with renewables contributing only a minor share of total production.

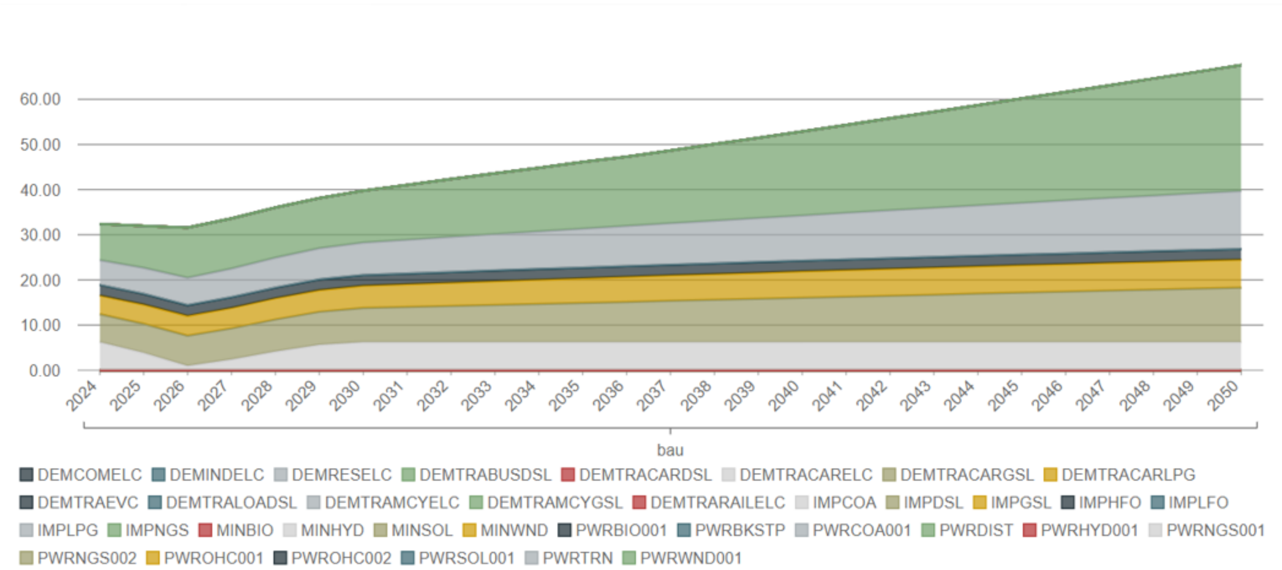


Figure 4: Annual technology emissions for the BAU scenario (2024–2050). Fossil technologies—especially oil, coal, and natural gas—drive a steady increase in total emissions due to rising electricity demand and minimal renewable deployment.

## 3.3 The Moderate Transition Path

Implementing the "Moderate Renewable Scenario" (Scenario 1) fundamentally alters the generation mix. Figure 5 breaks down the contribution of different renewable technologies.

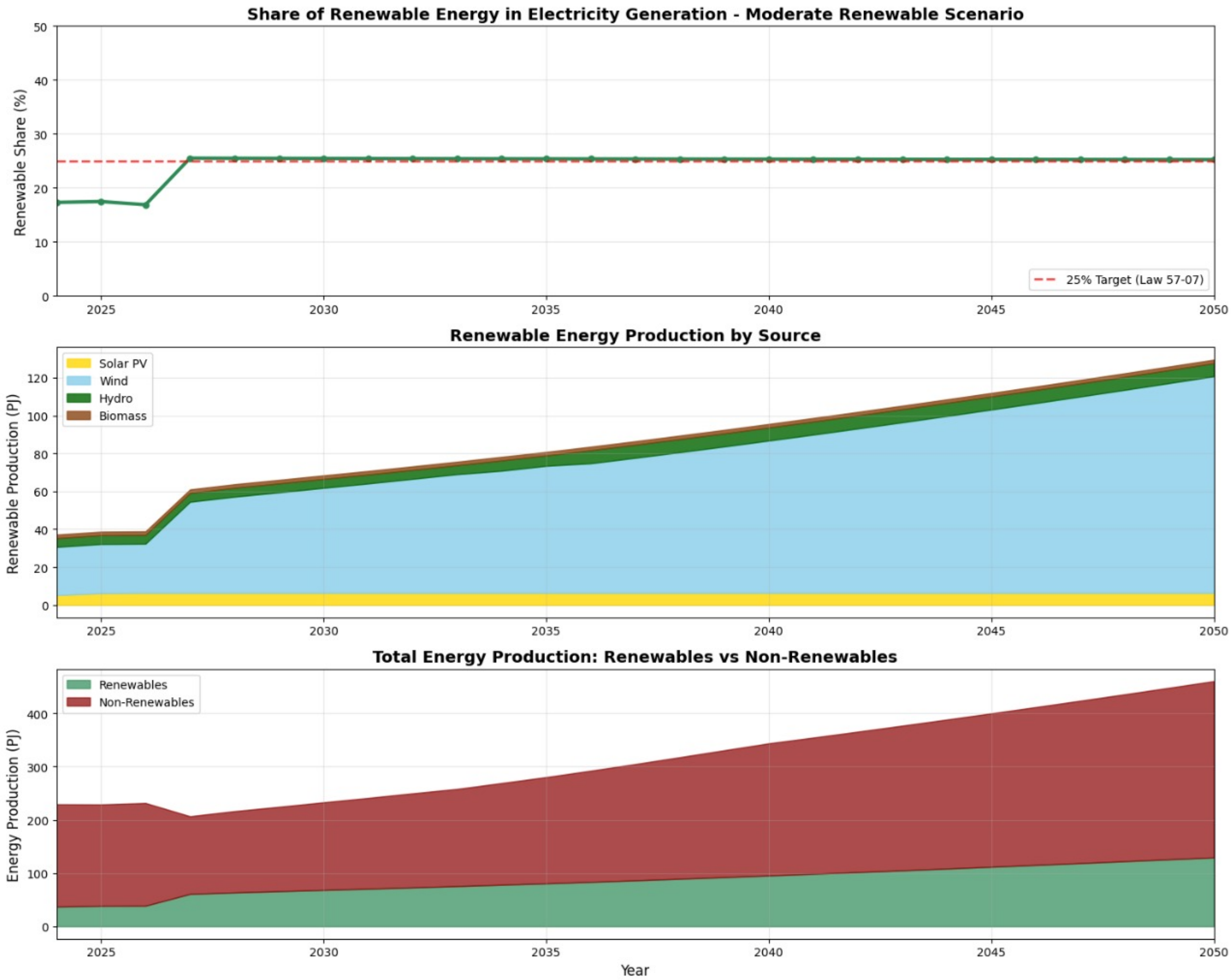


Figure 5: Moderate Renewable Scenario (Scenario 1) Dynamics. Top: Successful compliance with the 25% target. Middle: The "Wedge" of new energy is driven primarily by Wind (Light Blue) and Solar (Yellow). Bottom: Renewables (Green) begin to take a significant share of total energy production vs. Non-Renewables (Red).

The middle panel of Figure 5 is particularly instructive. It shows that while Solar PV provides a consistent base, *Wind Energy* (Light Blue area) acts as the primary driver for capacity expansion in the later years (2035-2050). This suggests that Wind is the cost-optimal technology for bulk renewable generation in the updated 2024 dataset.

### Scenario 1: Energy Production Trend

Figure 6 illustrates the annual electricity production by technology under Scenario 1. Unlike BAU, renewable technologies begin to capture a much larger share of total supply. Wind (WND) and Solar (SOL) expand steadily after 2030 due to the 25% renewable mandate imposed by Law 57-07. Hydro (HYD) remains constant, while fossil generation—especially LFO, HFO, and Diesel—declines progressively as they become less cost-competitive compared to new renewable installations.

A key observation is that **natural gas (NGS)** remains present but no longer dominates new generation additions. This shift indicates that the policy constraint successfully redirects the system toward cleaner technologies without requiring complete removal of fossil assets.

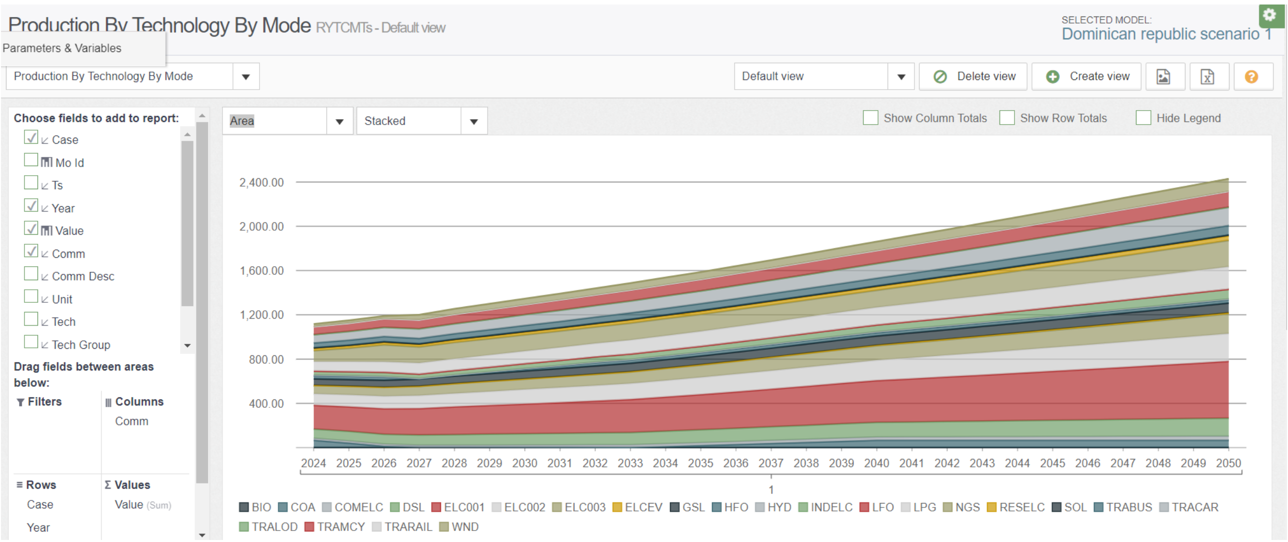


Figure 6: Annual electricity production by technology for Scenario 1 (Moderate Renewable Scenario). Wind and Solar show significant growth, driven by the 25% renewable energy requirement.

### Scenario 1: Emission Trajectory

Figure 7 shows the emissions pathway under Scenario 1. Total emissions drop substantially compared to BAU, especially after 2030, as wind and solar displace the most carbon-intensive fuels (HFO, LFO, and Diesel). The stacked area clearly shows the contraction of fossil-based emission sources over time.

The reduction, however, is not yet sufficient to meet the Paris Agreement 2030 target, indicating that while Scenario 1 is effective for renewable compliance, it is not aggressive enough for deep decarbonization.

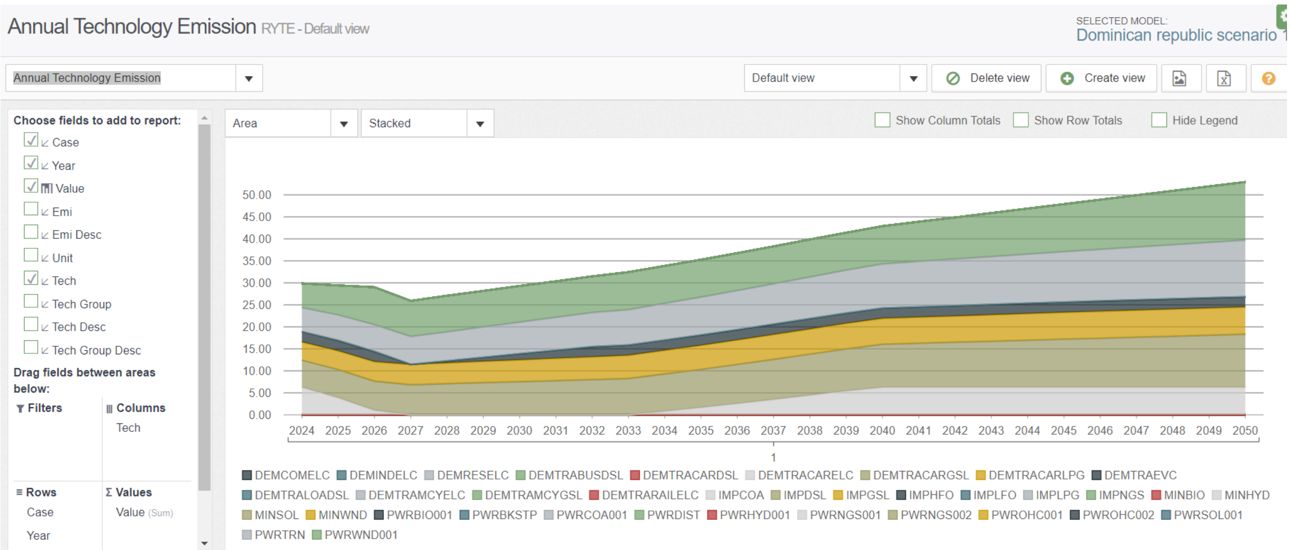


Figure 7: Annual emissions by technology for Scenario 1 (Moderate Renewable Scenario). Emissions decline steadily as wind and solar replace the most carbon-intensive fossil technologies.

# 4 Critical Analysis & Policy Implications

## 4.1 Unexpected Insights from Simulation Results

Contrary to initial expectations of a balanced renewable expansion, the OSeMOSYS optimization revealed several counter-intuitive dynamics:

1. **Wind Energy Dominance:** While policy discussions often emphasize solar PV due to its modularity, the model consistently selected wind energy as the primary driver for bulk renewable generation in cost-optimized scenarios (Scenario 1 & 2). This suggests that in the Dominican context, the superior capacity factors of wind resources outweigh the lower capital costs of solar PV for base-load displacement.
2. **The "Sticky" Nature of Natural Gas:** Despite aggressive renewable targets, natural gas capacity did not decline linearly. Instead, it remained a critical component of the grid mix through 2050. The model utilizes gas infrastructure not just for energy, but likely for grid flexibility—backing up the variable output of the massive wind and solar additions.
3. **Policy Efficacy Paradox:** Scenario 1 (Simple Target) achieved a slightly higher renewable penetration (∼30%) than Scenario 2 (Target + Mandated Capacity, ∼28%). This implies that prescriptive capacity mandates (forcing specific solar projects) may actually be *less* efficient than technology-neutral targets, as they prevent the model from fully exploiting the most cost-effective resource (wind).

## 4.2 Policy Recommendations

Based on these findings, we propose a phased policy roadmap for the Dominican energy sector:

### 4.2.1 Phase 1: Immediate Action (2024–2030)

- **Prioritize Wind Energy:** Shift investment focus and incentives toward wind power development, as it has emerged as the most cost-effective path to meeting the 25% target.
- **Secure Transition Fuel:** Acknowledge and plan for the continued role of Natural Gas as a bridge fuel. Infrastructure planning should account for its long-term necessity in a high-renewable grid.

### 4.2.2 Phase 2: Grid Maturity (2030–2040)

- **Flexibility First:** As renewable penetration exceeds 30%, priority must shift from simply "adding capacity" to "integrating capacity." This requires investment in grid modernization and utility-scale energy storage to manage the intermittency of the dominant wind/solar mix.
- **Coal Exit Strategy:** Develop a formal decommissioning plan for remaining coal assets, which the model identifies as economically obsolete under emission constraints.

### 4.2.3 Phase 3: Deep Decarbonization (2040–2050)

- **Advanced Technologies:** Explore green hydrogen and offshore wind to push beyond the 40-50% renewable ceiling.
- **Regional Interconnection:** Investigate subsea interconnections to improve grid stability and allow for the export of excess renewable generation.

# 5 Conclusion

The OSeMOSYS modeling confirms that meeting the Dominican Republic's energy targets is technically feasible but requires significant deviation from the current trajectory. The "Business As Usual" path fails to meet both national laws and international climate commitments. However, Scenarios 1 and 2 demonstrate that a mix heavily reliant on *Wind and Solar PV* can satisfy the 25% renewable mandate. The inclusion of the transport sector in this updated model highlights the scale of the challenge, as electrification of transport will require even greater renewable capacity additions than previously estimated.